

ACCORD: Dependency-Aware Conformal Risk Control for Hallucination-Resistant Multi-Agent Language Systems

Submitted by
Manikandan K
Neruban V N

Guided by
Nirmal Kumar V

Introduction

The Consensus Assumption & Its Failure

- Multi-agent LLMs (voting, debate, aggregation) assume *statistically independent* errors
- Shared datasets, reward models, & tokenizers → **correlated hallucinations**
- As correlation $\kappa \rightarrow 1$, consensus collapses to single-agent accuracy

ACCORD: Dependency-Aware Risk Control

- Gaussian Copula → Models inter-agent error correlation beyond naive Bayes
- Claim Dependency Graph (CDG) → Structured uncertainty propagation via belief propagation
- Conformal Risk Control (CRC) → Distribution-free, non-asymptotic hallucination guarantees
- POMDP Escalation Policy → Pareto-optimal accuracy–cost tradeoffs
- Heterogeneity Dividend → Empirical proof that structural diversity drives ensemble reliability

Problem Statement

The Broken Independence Assumption in Multi-Agent LLM Consensus

- **Standard Assumption:** Consensus mechanisms (majority voting, self-consistency, debate) require *statistically independent* agent errors to provide ensemble benefits
- **Real-World Violation:** Shared training datasets, RLHF reward models, tokenizer vocabularies, and prompt templates induce **correlated error distributions** across agents
- **Systemic Failure Mode:** As inter-agent correlation $\kappa \rightarrow 1$, majority voting collapses to single-agent accuracy—eliminating diversity benefits and enabling **correlated hallucinations** (confident, unanimous, wrong)
- **Critical Gap:** No existing framework explicitly models inter-agent error dependencies *or* provides distribution-free, non-asymptotic guarantees on hallucination rates for accepted claims
- **Consequence:** Deployed multi-agent systems silently produce high-confidence false outputs with no internal disagreement signal—posing undetectable risks in high-stakes applications

Motivation and Objective

MOTIVATION

- Multi-agent LLM consensus mechanisms (majority voting, self-consistency, debate) are now standard infrastructure for enterprise reasoning, yet they rest on a fundamentally flawed assumption: **agent errors are statistically independent**.
- In practice, shared training datasets, RLHF reward models, tokenizer vocabularies, and prompt templates induce highly correlated error distributions. As inter-agent correlation $\kappa \rightarrow 1$, consensus collapses to single-agent accuracy, eliminating ensemble diversity benefits entirely.
- This structural flaw produces **correlated hallucinations**—confident, unanimous, yet incorrect outputs that are invisible to existing evaluation metrics and undetectable by internal disagreement signals.
- Despite the growing deployment of agentic systems, no existing framework explicitly models inter-agent error dependencies or provides formal, distribution-free guarantees against this silent failure mode, creating a critical reliability gap for high-stakes applications.

Motivation and Objective

OBJECTIVE

- ACCORD aims to establish the first unified, theoretically grounded framework for hallucination-resistant multi-agent LLM reasoning under realistic correlated failure conditions. Specifically, it seeks to:
- **Model Correlated Failures:** Replace the independence assumption with a **Gaussian copula** that explicitly estimates inter-agent error correlation Σ and deflates confidence when agents share failure modes.
- **Propagate Structured Uncertainty:** Construct a learned **Claim Dependency Graph (CDG)** and apply loopy belief propagation to trace upstream logical errors, preventing isolated claim failures from cascading into systemic hallucinations.
- **Enforce Distribution-Free Guarantees:** Apply **Conformal Risk Control (CRC)** to compute a dynamic threshold λ^* that bounds the hallucination rate on accepted claims to a user-specified δ with confidence $1-\alpha$, regardless of the underlying data distribution.
- **Optimize Decision-Making:** Replace heuristic abstention rules with a **POMDP escalation policy** that Pareto-dominates fixed-threshold approaches on the accuracy–cost–latency frontier.
- **Quantify & Exploit Diversity:** Introduce **Correlated Hallucination Risk (CHR)** as a novel system-level metric, minimize it across benchmarks, and empirically characterize the **Heterogeneity Dividend** to provide principled guidelines for robust agent pool design.

Abstract

- **Problem:** Multi-agent LLM consensus mechanisms assume statistically independent agent errors—but shared training lineages and reward models induce **correlated hallucinations**, causing consensus to collapse under high agent correlation.
- **Solution:** We present **ACCORD**, a dependency-aware framework that:
 - Models inter-agent error correlations via a **Gaussian copula**
 - Propagates uncertainty through a learned **Claim Dependency Graph (CDG)**
 - Provides **distribution-free hallucination-rate guarantees** via Conformal Risk Control (CRC)
- **Results:** Across TruthfulQA, FEVER, and HotpotQA, ACCORD reduces **Correlated Hallucination Risk (CHR)** by up to **61%** over self-consistency baselines under high correlation, while maintaining competitive accuracy.
- **Key Insight:** ACCORD automatically identifies the **Heterogeneity Dividend**—exploiting diverse agent pools for reliable consensus and withholding output when homogeneity renders agreement unsafe.
- **Impact:** First unified framework with formal statistical guarantees for hallucination-resistant multi-agent LLM reasoning under realistic correlated failure conditions.

Literature Review

S.No	Title of the Paper	Journal / Conference Name	Author & Year	Methodology	Limitations
1	Self-Consistency Improves Chain of Thought Reasoning in Language Models	ICLR 2023	Wang et al., 2022 ResearchGate	Samples multiple CoT reasoning paths; majority vote aggregation	Assumes independent errors; same model generates all paths (maximum correlation $\kappa \rightarrow 1$); no formal guarantees
2	Improving Factuality and Reasoning in Language Models through Multiagent Debate	ICML / arXiv	Du et al., 2023 arXiv	Structured debate between multiple LLM agents; iterative refinement	Aggregates full responses (not claims); assumes independence; no distribution-free guarantees; no abstention mechanism
3	FACTScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation	EMNLP 2023	Min et al., 2023 arXiv	Decomposes text into atomic facts; retrieval-based verification using masked LM	Single-model evaluation; does not address multi-agent agreement; no risk control; no correlation modeling
4	SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative LLMs	EMNLP 2023	Manakul et al., 2023 arXiv	Sampling-based consistency check; detects hallucinations via response divergence	Single-model approach; no inter-agent dependency modeling; no formal hallucination rate guarantees
5	Conformal Risk Control	ICLR 2024	Angelopoulos et al., 2022 arXiv	Extends conformal prediction to control arbitrary monotone loss functions; distribution-free guarantees	Applied to single-model outputs; no multi-agent setting; no claim-level dependencies; no CDG structure

Literature Review

Multi-Agent Consensus Mechanisms

Self-Consistency (Wang et al., 2022)

- Proposed a decoding strategy that samples multiple chain-of-thought reasoning paths and takes majority vote ResearchGate
- Significantly improves reasoning accuracy over greedy decoding
- **Limitation:** Assumes statistically independent errors across samples; same model generates all paths (maximum correlation)

Multi-Agent Debate (Du et al., 2023)

- Introduced structured debate between multiple LLM agents to improve factuality and reasoning arXiv
- Shows performance gains with more agents and debate rounds ojs.aaai.org
- **Limitation:** Aggregates full responses rather than atomic claims; assumes independence; no formal guarantees

ChatEval (Chan et al., 2023)

- Multi-agent referee team that autonomously discusses and evaluates generated text quality openreview.net
- Uses role-based evaluation with multiple LLMs
- **Limitation:** Does not model inter-agent correlation structure; focused on evaluation, not generation Semantic Scholar

Literature Review

Hallucination Detection & Factual Evaluation

FactScore (Min et al., 2023)

- Fine-grained atomic evaluation metric that breaks generation into atomic facts aclanthology.org
- Computes percentage of atomic facts supported by knowledge source
- Uses retrieval and nonparametric masked LM for verification [arXiv](https://arxiv.org)
- **Limitation:** Evaluates individual outputs; does not address multi-agent agreement calibration

SelfCheckGPT (Manakul et al., 2023)

- Zero-resource, black-box hallucination detection using stochastic sampling consistency [arXiv](https://arxiv.org)
- Detects hallucinations by checking consistency across multiple sampled responses openreview.net
- **Limitation:** Single-model approach; does not model inter-agent dependencies; no formal risk guarantees
- **Key Gap:** Existing methods treat hallucination as generation-time phenomenon, not as correlated multi-agent failure

Literature Review

Conformal Prediction & Risk Control

Conformal Risk Control (Angelopoulos et al., 2022)

- Extends conformal prediction to control expected value of arbitrary monotone loss functions [arXiv](#)
- Provides distribution-free, finite-sample guarantees
- Enables bounding hallucination rate at user-specified threshold δ with confidence $1-\alpha$ [scholar.google.com](#)
- **Application to LLMs:** Prior work operates at token/sentence level for single models
- **Our Extension:** First to apply CRC to claim-level posteriors in multi-agent setting with correlation modeling

Theoretical Foundation:

- Guarantee: $P(R(\lambda^*) \leq \delta) \geq 1-\alpha$ holds for any exchangeable calibration set [GitHub](#)
- Minimum calibration size: $n = O(\log(1/\alpha)/\delta^2)$
- **Key Advantage:** No distributional assumptions beyond exchangeability

Literature Review

Dependency Modeling & Annotator Reliability

Dawid-Skene Model (1979)

- Classical EM algorithm for inferring ground truth from unreliable annotators ojs.aaai.org
- Estimates per-annotator error rates from observed labels
- Foundation for crowdsourcing and annotation aggregation ideas.repec.org
- **Limitation:** Original formulation assumes conditional independence given true label
- **Our Extension:** Generalizes D-S with copula-based correlation modeling + CDG + conformal constraints

Loopy Belief Propagation

- Applied to information extraction and relation detection on factor graphs
- **Our Use:** Propagates uncertainty through Claim Dependency Graph (CDG)
- Identifies upstream error sources in reasoning chains

POMDP-Based Decision Making

- Used for dialogue management and active information acquisition
- **Our Application:** Optimal escalation policy under partial observability of ground truth

Existing system – Its Drawback

Current Approaches

- **Majority Voting / Self-Consistency:** Aggregate outputs via simple majority or weighted vote across reasoning paths
- **Multi-Agent Debate:** Iterative refinement through structured agent discussion
- **LLM-as-Judge / RAG:** Single strong model or retrieval system validates outputs

Resulting Failure Mode

- Confident, unanimous, yet incorrect outputs with **no internal disagreement signal**
- Existing metrics (Accuracy, F1, ECE) remain high while **Correlated Hallucination Risk (CHR)** goes undetected

Existing system – Its Drawback

Drawback	Consequence
❓ Independence Assumption	Assumes agent errors are statistically independent; violated under shared training/reward models → <i>correlated hallucinations</i>
❓ No Dependency Modeling	Treats atomic claims as independent; ignores logical entailment → <i>upstream errors propagate undetected</i>
❓ No Formal Risk Guarantees	Heuristic thresholds only; no distribution-free bound on hallucination rate → <i>silent failure in high-stakes settings</i>
❓ Homogeneity Blindness	Cannot detect insufficient agent diversity; consensus confidence remains high even when $\kappa \rightarrow 1$ → <i>zero ensemble benefit</i>
❓ Claim-Agnostic Aggregation	Operates on full responses, not atomic claims → <i>cannot isolate or escalate specific factual errors</i>

Proposed System – Its Advantages

- **ACCORD**: *Dependency-Aware Conformal Risk Control for Hallucination-Resistant Multi-Agent LLM Systems*

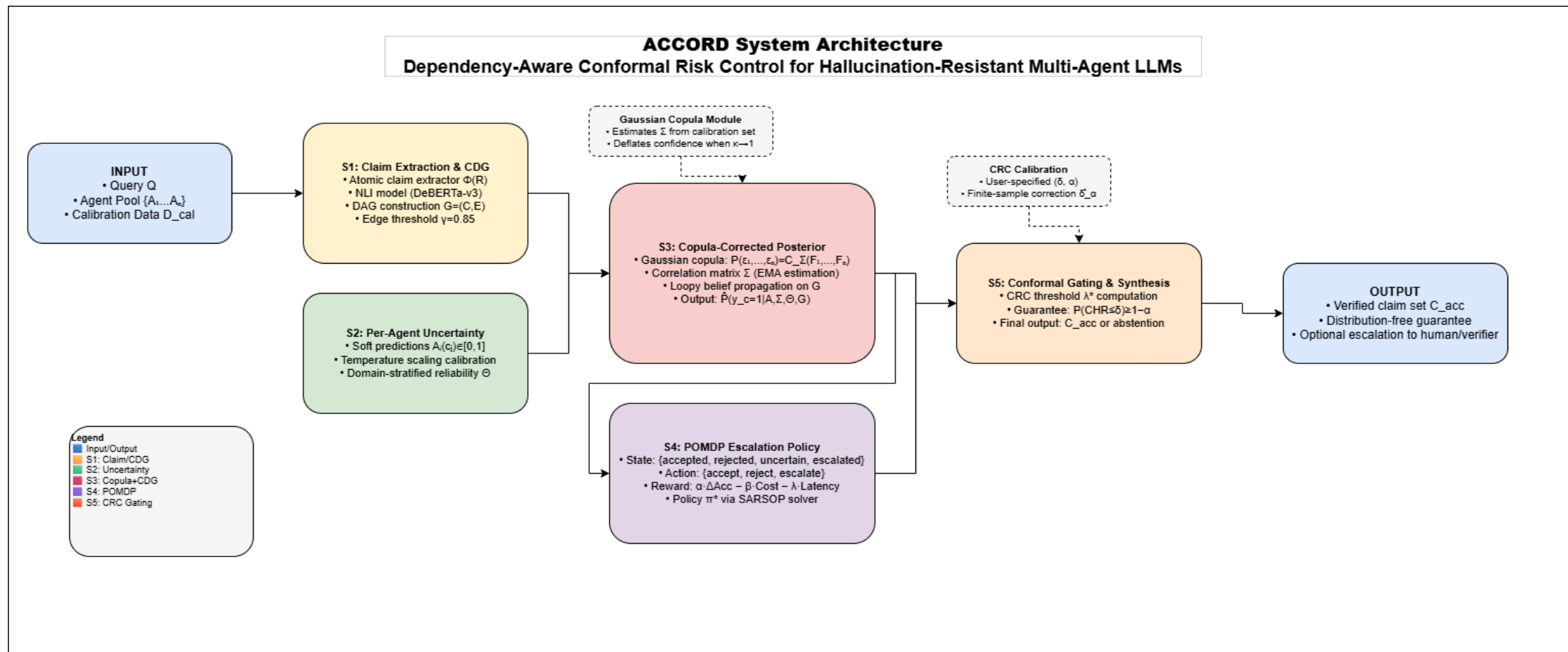
Empirical Impact

- Reduces **Correlated Hallucination Risk (CHR)** by up to **61%** vs. self-consistency baselines (TruthfulQA, FEVER, HotpotQA)
- Maintains competitive accuracy while providing **formal statistical guarantees**
- First unified framework for **risk-controlled, dependency-aware multi-agent reasoning** under realistic correlated failure conditions

Proposed System – Its Advantages

Advantage	Technical Benefit
❓ Gaussian Copula Modeling	Explicitly captures inter-agent error correlations (Σ); deflates confidence when agents share failure modes—prevents overconfident consensus under $\kappa \rightarrow 1$
❓ Claim Dependency Graph (CDG)	Propagates uncertainty via loopy belief propagation; isolates upstream error sources in reasoning chains instead of treating claims independently
❓ Conformal Risk Control (CRC)	Provides distribution-free, non-asymptotic guarantee : $P(\text{Hallucination Rate} \leq \delta) \geq 1 - \alpha$ on accepted claims—no assumptions beyond exchangeability
❓ POMDP Escalation Policy	Pareto-optimal accuracy–cost–latency tradeoffs; dynamically chooses accept/reject/escalate based on belief state, outperforming fixed-threshold heuristics
❓ Heterogeneity Dividend	Automatically identifies when agent diversity enables reliable consensus vs. when homogeneity warrants abstention—enables principled agent pool design

System Architecture



Modules

Modules Explanation

Results and Discussion

Conclusion

Thank You